

Discussion 6- Solutions

1. Suppose you are playing a dart game and the probability of hitting the bullseye on any given throw is 0.40 and the throws are independent.

- (a) What is the probability that you would get three bullseyes in nine throws?
- (b) What is the probability that you would get at least six bullseyes in nine throws?

Each throw is one trial and throws are independent.

Success= Hitting the bull's eyes

$P(\text{Hitting the bulls eyes in one throw}) = \text{Probability}(\text{success in one trail}) = 0.4.$

(a) Let W = number of success(hitting bulls eyes) in $n = 9$ throws(trials). $W \sim \text{Bin}(n = 9, p = 0.4)$

$$P(W = 3) = \binom{9}{3} (1 - p)^6 p^3 = \frac{9!}{6!3!} (1 - 0.4)^6 \cdot (0.4)^3 = \frac{9 \cdot 8 \cdot 7}{6} \cdot (0.6)^6 \cdot (0.4)^3 = 84 \cdot (0.6)^6 \cdot (0.4)^3$$

(b) $W \sim \text{Bin}(n = 9, p = 0.4).$

$$\begin{aligned} P(W \geq 6) &= \sum_{k=6}^9 P(W = k) = \sum_{k=6}^9 \frac{9!}{k!(9-k)!} (0.6)^{9-k} \cdot (0.4)^k = \\ &= \frac{9!}{6!3!} (0.6)^3 \cdot (0.4)^6 + \frac{9!}{7!2!} (0.6)^2 \cdot (0.4)^7 + \frac{9!}{8!(9-8)!} (0.6) \cdot (0.4)^8 + \frac{9!}{9!(0)!} (0.4)^9 \\ &= 84 \cdot (0.6)^3 \cdot (0.4)^6 + 36 \cdot (0.6)^2 \cdot (0.4)^7 + 9 \cdot (0.6) \cdot (0.4)^8 + (0.4)^9 \\ &= (0.6) \cdot (0.4)^6 (84 \cdot (0.36) + 36 \cdot (0.24) + 9 \cdot (0.16) + 0.064) \\ &= (0.6) \cdot (0.4)^6 (30.24 + 8.64 + 1.44 + 0.064) \\ &= (0.6) \cdot (0.4)^6 (40.384) = 24.2304 \cdot (0.4)^6 = 0.09924772 \end{aligned}$$

Alternatively, for this part one could use R to do the above calculation. If $X \sim \text{Bin}(n, p)$ and $F(a)$ shows the cdf at $X = a$, you can use R as a calculator, the command `pbinom(a, size=n, prob=p)` gives $F(a)$. For the above

$P(W \geq 6) = 1 - P(W \leq 5) = 1 - F(5)$, In R type `1 - pbinom(5, size=9, prob=0.4)` the output is 0.09935258, which is very close to the value we obtained above.

2. Suppose a survey found that approximately 30% of students believe the quality of food in the campus dining hall is unsatisfactory. Let X represent the number of students in a random sample of $n = 50$ who hold this belief. Assuming the survey results are still accurate, find the following probabilities using cdf tables.

- (a) X is at most 25.
- (b) X is equal to 14.

(c) X is at least 31.

Problem 2 Solution. Success = Believing quality of food is unsatisfactory

$$P(\text{Success}) = 0.3$$

$$X \sim \text{Bin}(n = 50, p = 0.3)$$

If we use F for cdf, then

(a) $P(X \leq 25) = F(25) = 0.9990668$ in R use `pbinom(25, size=50, prob=0.3)`

(b) $P(X = 14) = F(14) - F(13) = 0.4468316 - 0.3278832 = 0.1189483$ for the exact probability

In R use `dbinom(14, size=50, prob=0.3)` for the exact value of Pdf at a given point for Binomial distribution.

(c) $P(X \geq 31) = 1 - P(X < 31) = 1 - P(X \leq 30) = 1 - F(30) = 1 - 0.9999972 = 2.8467 \times 10^{-6}$.

In R use `1 - pbinom(30, size=50, prob=0.3)`

3. A random variable X has a binomial distribution with mean 5 and variance 2.5. Find $P(X=2)$.

Problem 3 Solution. $X \sim \text{Bin}(n, p)$.

$$np = 5, np(1 - p) = 2.5$$

$$5(1 - p) = 2.5$$

$$p = 1 - \frac{2.5}{5} = 0.5$$

$$n = \frac{5}{0.5} = 10$$

Hence, $X \sim \text{Bin}(10, 0.5)$.

$$P(X = 2) = \binom{10}{2} (0.5)^2 (0.5)^8 = 45(0.5)^{10}.$$

4. Sarah works as a sales representative selling house insurance over the phone. Only 5% of all calls result in a successful sale. Assume that the outcome of each call is independent of the others.

(a) If Sarah makes 10 calls, what is the probability that she makes exactly 3 sales?

(b) If Sarah makes 10 calls, what is the probability that she makes at least 2 sales?

(c) If Sarah makes 10 calls, what is the probability that she makes at most 2 sales?

Problem 4 Solution. probability of sale on each call = $P(\text{success}) = 0.05$

(a) X = number of success (sales) in 10 trials. $X \sim \text{Bin}(10, 0.05)$

$$P(X = 3) = \binom{10}{3} (0.05)^3 (0.95)^7$$

(b)

$$\begin{aligned} P(X \geq 2) &= 1 - (P(X = 1) + P(X = 0)) = 1 - ((0.95)^{10} + 10 \cdot 0.05 \cdot (0.95)^9) \\ &= 1 - (0.95)^9 (0.95 + 0.5) = 1 - (0.95)^9 (1.45) \end{aligned}$$

(c)

$$\begin{aligned} P(X \leq 2) &= P(X = 2) + P(X = 1) + P(X = 0) = \binom{10}{2} (0.05)^2 (0.95)^8 + (0.95)^9 (1.45) \\ &= 45 \cdot 0.0025 \cdot 0.95^8 (0.95)^9 (1.45) = 0.95^8 (45 \cdot 0.0025 + 0.95 \cdot 1.45) = 0.95^8 (0.1125 + 1.3775) \\ &= 0.95^8 (1.49) = 0.9885 \end{aligned}$$

5. Suppose we have two coins. Coin C_1 comes up heads with probability 0.2 and coin C_2 comes up heads with probability 0.7. We repeat this process 4 times:

- Choose a coin with equal probability.
- Flip that coin once.

Suppose X is the number of heads after 4 flips.

- What is $E[X]$?
- What is $\text{Var}(X)$?
- Based on the number of heads we get, we earn $Y = \frac{2}{X+1}$. What's $E[Y]$?

Solutions. Let X be the number of heads. At each trial,

$$\begin{aligned} P(H) &= P(H|C_1)P(C_1) + P(H|C_2)P(C_2) \\ &= (0.2) \cdot (0.5) + (0.7) \cdot (0.5) \\ &= 0.1 + 0.35 = 0.45 \end{aligned}$$

Method 1.

(a) Now we can define $X_i = \begin{cases} 1 & \text{if the } i^{\text{th}} \text{ trial is head} \\ 0 & \text{if the } i^{\text{th}} \text{ trial is tail} \end{cases}$

each $X_i \sim \text{Bern}(p = P(H))$. Hence $E[X_i] = 0 \cdot P(X_i = 0) + 1 \cdot P(X_i = 1) = P(H) = 0.45$
 Note that $X = X_1 + X_2 + X_3 + X_4$. Hence, by linearity of expectation

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4] = 4 \cdot (0.45) = 1.8$$

(b) Since $X = X_1 + X_2 + X_3 + X_4$ where X_i 's are independent, we have

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) + \text{Var}(X_4) = 4 \cdot (0.45) \cdot (0.55) = 0.99$$

Second method:

$$X \sim \text{Bin}(n = 4, p = 0.45)$$

$$E[X] = np = 1.8, \text{Var}(X) = np(1 - p) = 0.99.$$

$$P(X = k) = \binom{4}{k} (0.45)^k (0.55)^{4-k} \quad ; k = 0, 1, 2, 3, 4.$$

$$\begin{aligned} E[Y] &= E\left[\frac{2}{X+1}\right] = \sum_{i=0}^4 \frac{2}{k+1} P(X = k) \\ &= \sum_{i=0}^4 \frac{2}{k+1} \binom{4}{k} (0.45)^k (0.55)^{4-k} \end{aligned}$$